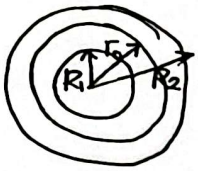
15.3



当施加的电压为U_0时，

~~内层介质受到的最大场强 E_{ε_1}~~

~~由$\vec{D}$的高斯定律有~~

$\vec{D} = \varepsilon_0 \varepsilon_1 \vec{E}$

设内筒单位长度带电量为λ

则 $D = \frac{\lambda}{2\pi R}$ （R为高斯面圆柱的半径）

内层介质中的电场强度

$E_1 = \frac{\lambda}{2\pi\varepsilon_0\varepsilon_1 r}$ $(R_1 \leqslant r \leqslant r_0)$ 则 $E_{1\,max} = \frac{\lambda}{2\pi\varepsilon_0\varepsilon_1 R_1}$

外层介质中的电场强度

$E_2 = \frac{\lambda}{2\pi\varepsilon_0\varepsilon_2 r}$ $(r_0 \leqslant r \leqslant R_2)$ 则 $E_{2\,max} = \frac{\lambda}{2\pi\varepsilon_0\varepsilon_1 r_0}$

即 $\frac{E_{1\,max}}{E_{2\,max}} = \frac{\varepsilon_2 r_0}{\varepsilon_1 R_1} = \frac{r_0}{2R_1} < \frac{R_2}{2R_1} < 1$

即 $E_{1\,max} < E_{2\,max}$ 故外层介质先被击穿

当 $E_{2\,max} = E_{max}$ 时，$\frac{\lambda}{2\pi\varepsilon_0\varepsilon_2 r_0} = E_{max}$

则 $E_1 = \frac{r_0}{2r} E_{max}$，$E_2 = \frac{r_0}{r} E_{max}$

两筒间的电势差 $U = \int_{R_1}^{r_0} E_1 dr + \int_{r_0}^{R_2} E_2 dr$

$= \frac{r_0}{2} E_{max} \ln\frac{r_0}{R_1} + r_0 E_{max} \ln\frac{R_2}{r_0}$

$= \frac{r_0 E_{max}}{2} \ln\frac{R_2^2}{R_1 r_0}$

15.6 (1)对于各向同性线性电介质中，电位移与电场满足

$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$

由$\vec{D}$的高斯定律有：

$\oint_S \vec{D} d\vec{S} = q = \sigma S$

又电位移通量与高斯面垂直 则 $\oint_S \vec{D} d\vec{S} = \iint_S D dS = D \cdot S$ 可知 $D = \sigma$

故 $E = \frac{D}{\varepsilon_0 \varepsilon_r} = \frac{\sigma}{\varepsilon_0 \varepsilon_r} = \frac{0.52 \times 10^{-3}}{8.85 \times 10^{-12} \times 6.0}$ N/C $= 9.79 \times 10^6$ N/C

方向由内表面指向外表面

(2) $U = E \cdot d = 9.79 \times 10^6 \times 5.2 \times 10^{-9}$ V

$= 5.09 \times 10^{-2}$ V

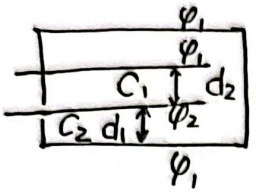
扫描全能王 创建

15.14 1° 板和盒子未用导线连接时

金属盒不影响平行板间电场

$$C=\frac{\varepsilon_0 S}{d_2}=\frac{8.85\times10^{-12}\times0.02}{0.5\times10^{-3}}=3.54\times10^{-10}F$$

2° 板和盒子用导线连接时，该板与盒子间无电场



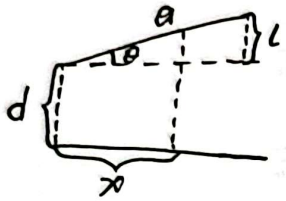
此时电容器相当于两个电容并联

$$C_1=\frac{\varepsilon_0 S}{d_2}=3.54\times10^{-10}F$$

$$C_2=\frac{\varepsilon_0 S}{d_1}=\frac{2\varepsilon_0 S}{d_1}=7.08\times10^{-10}F$$

$$C=C_1+C_2=1.062\times10^{-9}F$$

15.15



由 $C=\frac{\varepsilon_0 S}{d_0}$ 可知 $dC=d\left(\frac{\varepsilon_0 S}{d_0}\right)=\frac{\varepsilon_0}{d_0(x)}dS$

其中 $S=b\cdot x$

$d_0(x)=d+\frac{x}{a}l$ 则 $dC=\frac{\varepsilon_0\cdot b}{d+\frac{x}{a}l}dx$

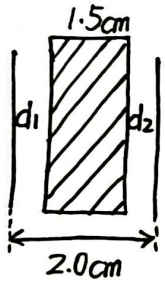
$$C=\int dC=\int_0^a\frac{\varepsilon_0 b}{d+\frac{x}{a}l}dx=\varepsilon_0 b\int_0^a\frac{dx}{d+\frac{x}{a}l}=\frac{\varepsilon_0 ba}{l}\int_0^a\frac{d\frac{xl}{a}}{d+\frac{xl}{a}}$$

$$=\frac{\varepsilon_0 ba}{l}\int_0^l\frac{dt}{d+t}=\frac{\varepsilon_0 ba}{l}\ln\frac{d+l}{d}=\frac{\varepsilon_0 ab}{l}\ln\left(1+\frac{l}{d}\right)$$

又 $l\ll d$ 则 $\ln\left(1+\frac{l}{d}\right)=\frac{l}{d}-\frac{l^2}{2d^2}$

故 $C=\frac{\varepsilon_0 ab}{l}\cdot\frac{l}{d}\left(1-\frac{l}{2d}\right)=\frac{\varepsilon_0 ab}{d}\left(1-\frac{l}{2d}\right)$ （近似于 $\frac{\varepsilon_0 ab}{d}$）

15.16



插入过程中，电容器电荷不变

$C_0U_0=C'U'$ 其中 $\frac{U'}{U_0}=0.6$ 解 $C'=\frac{5}{3}C_0$

插入平板材料后，电容器相当于三个电容器串联

$C_1=\frac{\varepsilon_0 S}{d_1}$，$C_2=\frac{\varepsilon_0 S}{d_2}$，$C_3=\frac{\varepsilon_1 S}{d_3}$ 且 $d_1+d_2+d_3=d=2.0cm$

$d_3=1.5cm$

$$\frac{1}{C'}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}=\frac{d_3}{\varepsilon_1 S}+\frac{d_1+d_2}{\varepsilon_0 S}$$

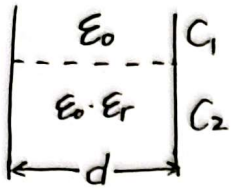
又 $C_0=\frac{\varepsilon_0 S}{d}$ 则 $\frac{1}{C'}=\frac{3d}{5\varepsilon_0 S}$

解 $\varepsilon_1=\frac{1}{\frac{\frac{3}{5}\times2-0.5}{1.5}}\varepsilon_0=\frac{15}{7}\varepsilon_0=2.14\varepsilon_0$

即该电介质的相对介电常量为 2.14

扫描全能王 创建

15.20



电容器可视为上下两个电容器并联

$$C = C_1 + C_2$$

$$C_1 = \frac{\varepsilon_0 S_1}{d} = \frac{\varepsilon_0 (a-h) b}{d}$$

$$C_2 = \frac{\varepsilon_0 \varepsilon_r S_2}{d} = \frac{\varepsilon_0 \varepsilon_r h b}{d}$$ 其中b, d均为常量

$$C = \frac{\varepsilon_0 \varepsilon_{r,eff} S_3}{d} = \frac{\varepsilon_0 \varepsilon_{r,eff} a b}{d}$$

联立可知 $a - h + \varepsilon_r h = \varepsilon_{r,eff} a$

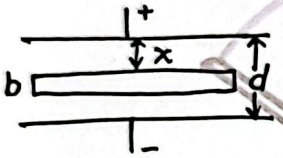
解 $\varepsilon_{r,eff} = \frac{\varepsilon_r \cdot h - h + a}{a} = \frac{(\varepsilon_r - 1)h}{a} + 1$

当 $\varepsilon_r = 1.95$, $\frac{\varepsilon_r - 1}{a} = \frac{0.95}{a}$

当 $\varepsilon_r = 33$, $\frac{\varepsilon_r - 1}{a} = \frac{32}{a}$ 明显大于 $\frac{0.95}{a}$

故此种油量计对甲醇更敏感，甲醇燃料更适宜此油量计

15.24 (1)



未插入金属板时，$C_0 = \frac{\varepsilon_0 S}{d}$

插入金属板后，总电容相当于两电容串联

$C_1 = \frac{\varepsilon_0 S}{x}$, $C_2 = \frac{\varepsilon_0 S}{d-b-x}$

$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2}$ 解 $C' = \frac{\varepsilon_0 S}{d-b}$

又电量Q不变

则电容器储能增加 $\Delta E = \frac{1}{2}\frac{Q^2}{C'} - \frac{1}{2}\frac{Q^2}{C_0}$

$= \frac{-Q^2 \cdot b}{2\varepsilon_0 S}$ （即储能减少 $\frac{Q^2 b}{2\varepsilon_0 S}$）

(2) 电容器储能减少，则电场力对金属板做正功

$W_{电} = -\Delta E$

又金属板总能量不变

$W_{外} + W_{电} = 0$ 解 $W_{外} = -\frac{Q^2 b}{2\varepsilon_0 S}$

外力对其做负功，则其是被吸入进两极板中

(3) 保持电压U不变则有 $C_0 Q = C' Q'$ 且 $Q = C_0 U$

$\Delta E' = \frac{1}{2}Q'U - \frac{1}{2}QU = \frac{U}{2} \cdot \frac{\varepsilon_0 S b}{d(d-b)} \cdot U = \frac{U^2 \varepsilon_0 S b}{2d(d-b)}$ （储能增加）

电源对电容器做功

$W_E = \Delta Q U = (Q' - Q)U = 2\Delta E'$

由能量守恒有

$W_{外} + W_E = \Delta E'$ 解 $W_{外} = -\Delta E' = -\frac{U^2 \varepsilon_0 S b}{2d(d-b)}$

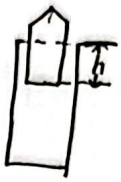
外力对其做负功，则其是被吸入两极板中

扫描全能王 创建

15.25

$C=\frac{\varepsilon_0 S}{r}$

$dC=\frac{\varepsilon_0 dS}{r}=\frac{\varepsilon_0 2\pi h dr}{r}$



则 $C=\int dC=\int_{R_1}^{R_2}\frac{\varepsilon_0 2\pi h dr}{r}=\frac{2\pi\varepsilon_0 h}{\ln\frac{R_2}{R_1}}$

电源撤除后，电容器电荷量Q不变

$Q=CU$

电容器储能

$E=\frac{1}{2}QU=\frac{1}{2}CU^2=\frac{2\pi\varepsilon_0 U^2 h}{2\ln\frac{R_2}{R_1}}$

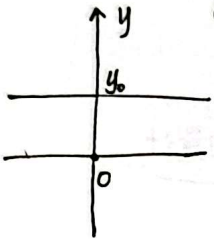
则 $dE=\frac{\pi\varepsilon_0 U^2 dh}{\ln\frac{R_2}{R_1}}$

电场力做功 $W_{电}=-\Delta E$ 即 $dW_{电}=-dE$

$F=\frac{dW_{电}}{-dh}=\frac{-dE}{-dh}=\frac{\pi\varepsilon_0 U^2}{\ln\frac{R_2}{R_1}}=\frac{3.14\times8.85\times10^{-12}\times(5\times10^3)^2}{\ln\frac{7}{5}}N$

$=2.06\times10^{-3}N$ 方向向下

15.28



(1) $\varepsilon_r=1+\frac{3}{y_0}y$ 取 dy 长度的薄层，其等效电容dC

$C=\frac{\varepsilon_0\varepsilon_r S}{y}$ 则 $dC=\frac{\varepsilon_0\varepsilon_r S}{dy}$

又电容器串联 则 $\frac{1}{C}=\int\frac{1}{dC}=\int_0^{y_0}\frac{dy}{\varepsilon_0\varepsilon_r S}=\frac{1}{\varepsilon_0 S}\int_0^{y_0}\frac{dy}{1+\frac{3}{y_0}y}$

$=\frac{y_0}{3\varepsilon_0 S}\ln\frac{4}{1}=\frac{2\ln2\, y_0}{3\varepsilon_0 S}$

$C=\frac{3\varepsilon_0 S}{2\ln2\cdot y_0}$

(2) 面束缚电荷密度 $\sigma'=\vec{P}\cdot\vec{e}$

$=\varepsilon_0(\varepsilon_r-1)\vec{E}\cdot\vec{e}$

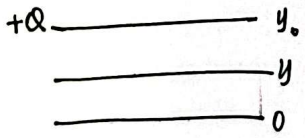
上表面

$\varepsilon_r=1$ 则 $\sigma_1'=0$ （平行板电容器 $D=\sigma_0$）

下表面

$\varepsilon_r=1+3=4$ 则 $\sigma_2'=-3\varepsilon_0 E=-3\varepsilon_0\cdot\frac{D}{\varepsilon_0\varepsilon_r}=-\frac{3}{4}D=-\frac{3}{4}\sigma_0$

(3) 取平行于平行板的一平面作为高斯面，对高为dy的封闭面使用高斯定律 $=-\frac{3Q}{4S}$



$dq_{in}=\rho\, dy S_0$ （ρ为在y处的面束缚电荷密度）

又 $dq_{in}=dD\cdot S_0=\varepsilon_0\cdot S_0\cdot dE$

$=\varepsilon_0\cdot\cdot S_0\cdot d\frac{D}{\varepsilon_0\varepsilon_r}$

$=S_0\cdot\frac{Q}{S}\cdot d\frac{1}{\varepsilon_r}$

故 $\rho=\frac{Q}{S}\frac{d\frac{1}{\varepsilon_r}}{dy}=\frac{Q}{S}\left[\left(1+\frac{3}{y_0}y\right)^{-1}\right]'=\frac{3Q}{4S}\left(1+\frac{3}{y_0}y\right)^{-2}$

整个介质体积内的总束缚电荷

$Q'=-\int_0^{y_0}\rho S\cdot dy=-\int_0^{y_0}Q\left[\left(1+\frac{3}{y_0}y\right)^{-1}\right]'dy=-Q\left(1+\frac{3}{y_0}y\right)^{-1}\Big|_0^{y_0}$

$=\frac{3}{4}Q$

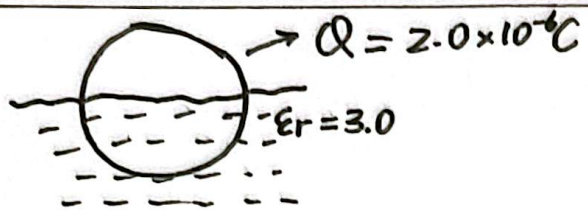
(4) 表面的面束缚电荷

$Q_{表}=\sigma_1'S+\sigma_2'S=-\frac{3}{4}Q$ 则 $Q'+Q_{表}=0$

故体束缚电荷加上面束缚电荷，其总和为0

扫描全能王 创建

15.29.



设上、下部分所带电量分别为 Q_1, Q_2

则 $Q=Q_1+Q_2=2.0\times10^{-6}C$

$$E=\frac{\sigma}{\varepsilon_0\varepsilon_r}$$

$$\varphi=ER=\frac{\sigma R}{\varepsilon_0\varepsilon_r}$$

上半部分电势 $\varphi_1=\dfrac{\sigma_1 R}{\varepsilon_0\varepsilon_{r_1}}=\dfrac{R}{\varepsilon_0}\cdot\dfrac{Q_1}{2\pi R^2}$

下半部分电势 $\varphi_2=\dfrac{\sigma_2 R}{\varepsilon_0\varepsilon_{r_2}}=\dfrac{R}{3\varepsilon_0}\dfrac{Q_2}{2\pi R^2}$

又 $\varphi_1=\varphi_2$ 则 $Q_2=3Q_1$

又 $Q_1+Q_2=2.0\times10^{-6}C$

解 $\begin{cases} Q_1=5\times10^{-7}C \\ Q_2=1.5\times10^{-6}C \end{cases}$

扫描全能王 创建